LARYNX

First part

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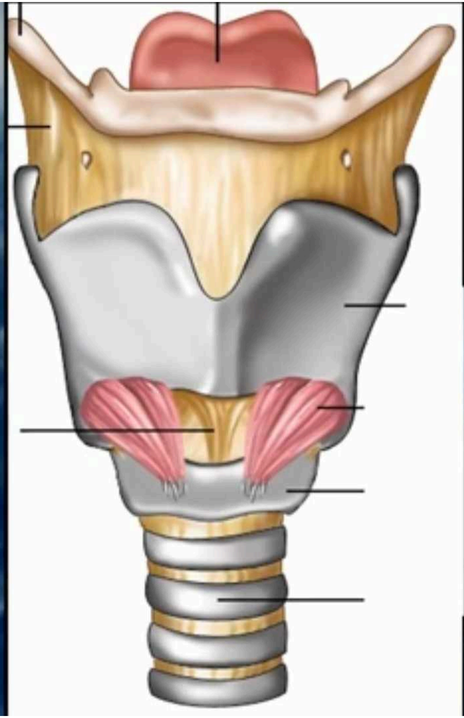
Department of Anatomy



LARYNX

First part:

- Introduction (shape , length , extend function).
- Surface marking and over view about larynx.
- Relation.
- Structure (components).

  1- Cartilaginous skeleton.

  2- Membranes and ligaments.

  3- joints.

  4- Muscles (Intrinsic & extrinsic muscles).

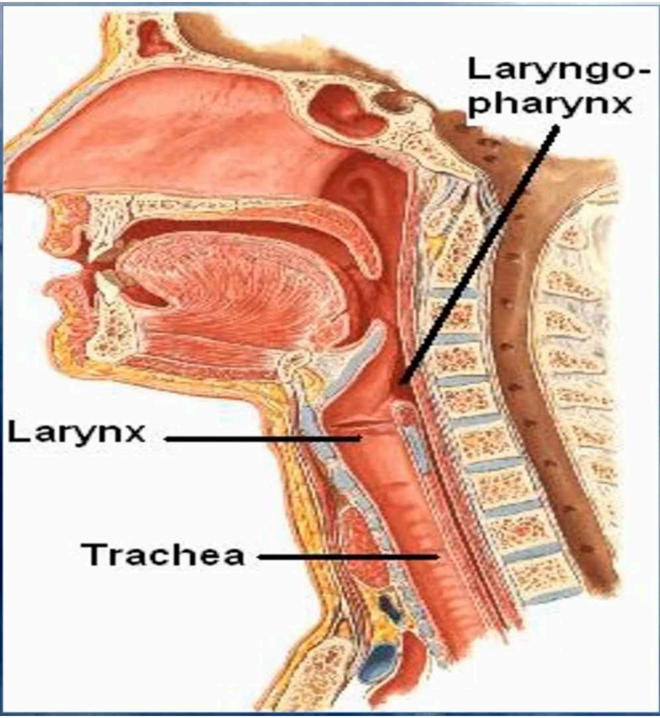
  5- Mucosal lining.

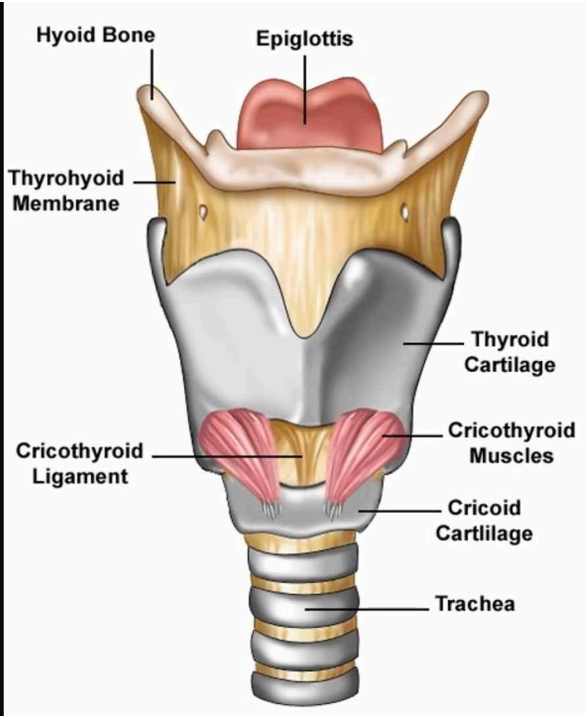
Secod part

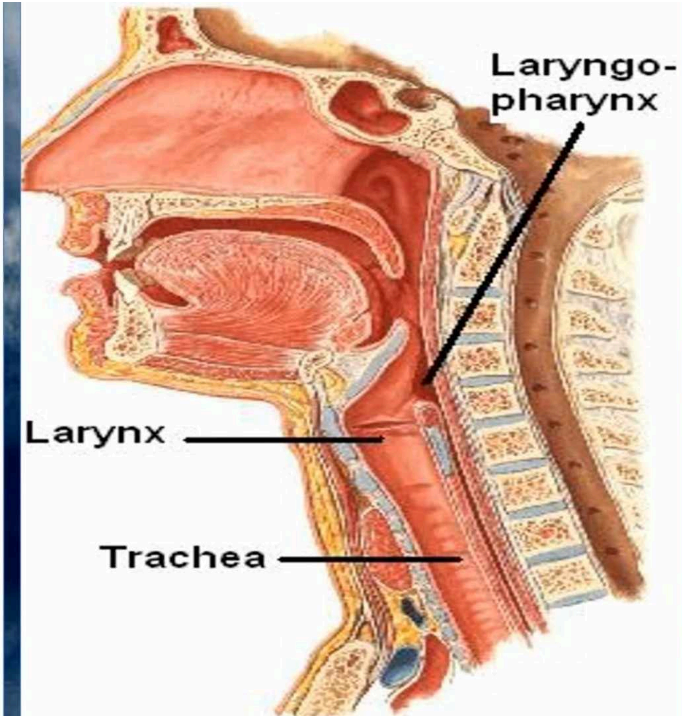
- Laryngeal Inlet.
- Laryngeal Cavity.
- Blood Supply.
- Nerve Supply.
- Clinical note.

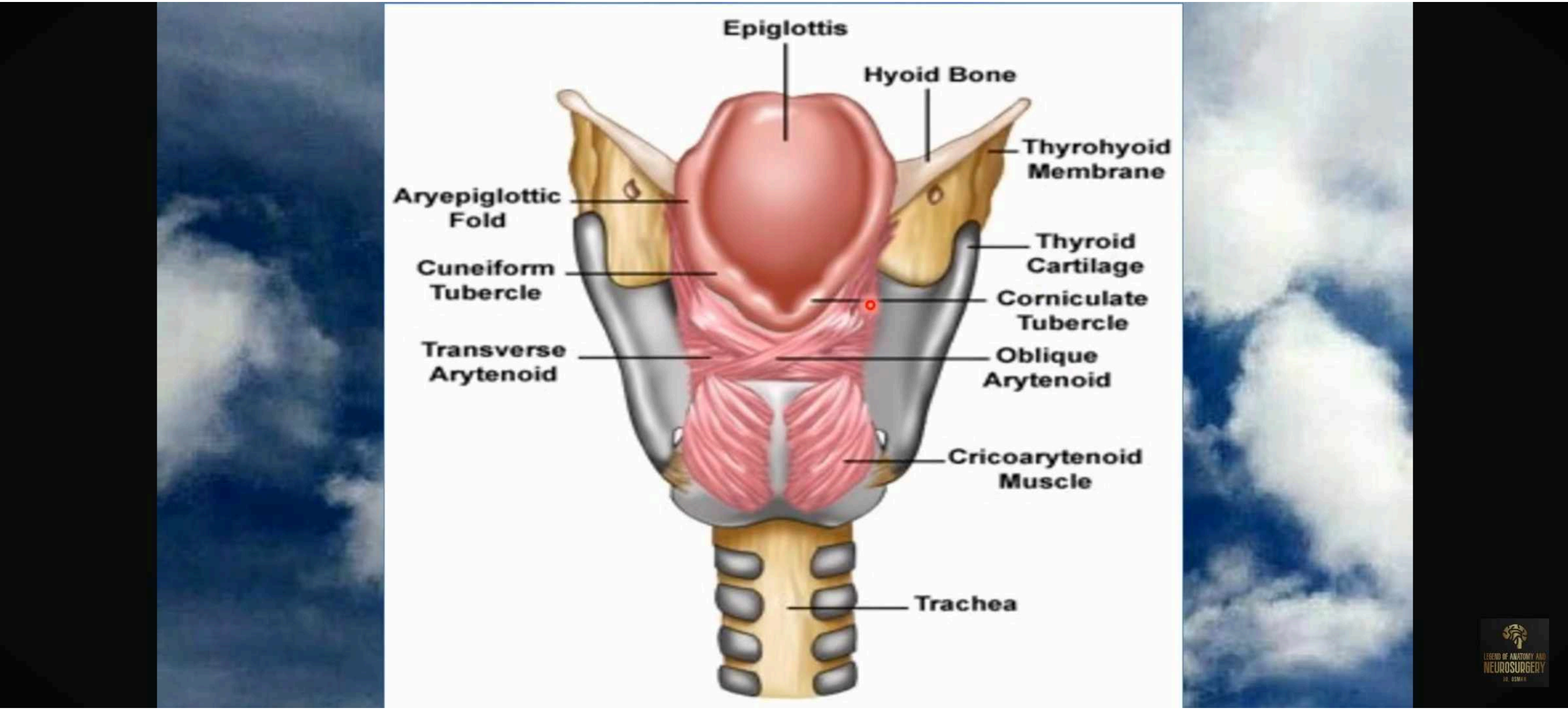
LARYNX

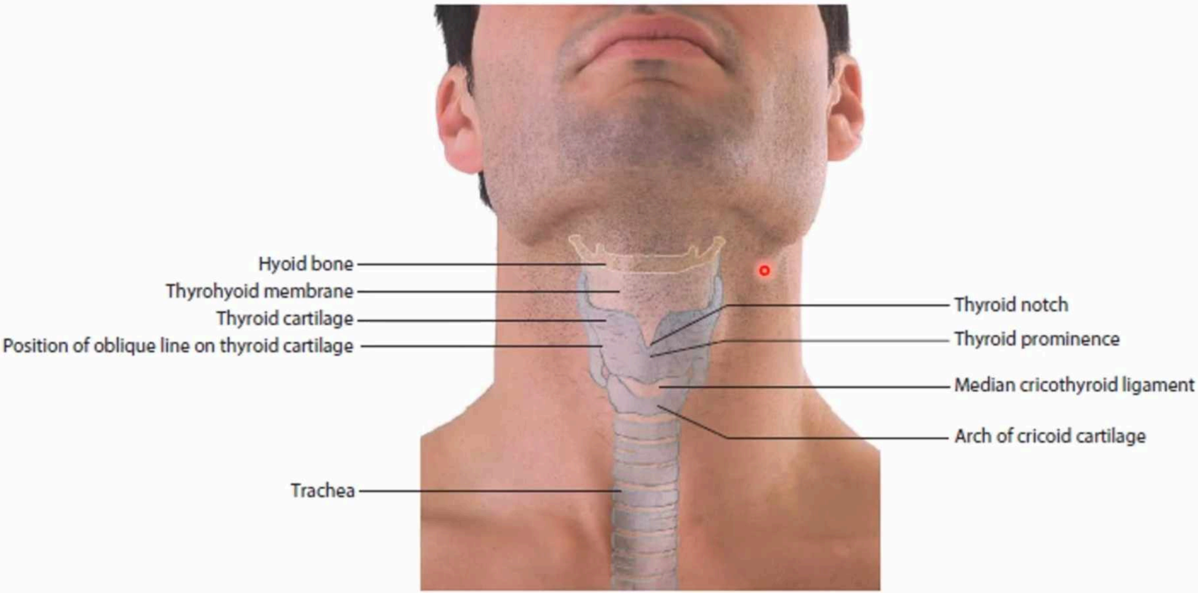
- The larynx is the part of **the respiratory tract** which contains **the vocal cords.**
- In adult it is **2-inch-long** tube.
- It opens **above** into the laryngeal part of the **pharynx.**
- **Below,** it is continuous with the trachea
- The larynx lies below the hyoid bone in the midline of the neck at the level of C4–6 vertebrae.
- **The larynx has functions in**:
  - Respiration (breathing).
  - Phonation (voice production).
  - Deglutition (swallowing).



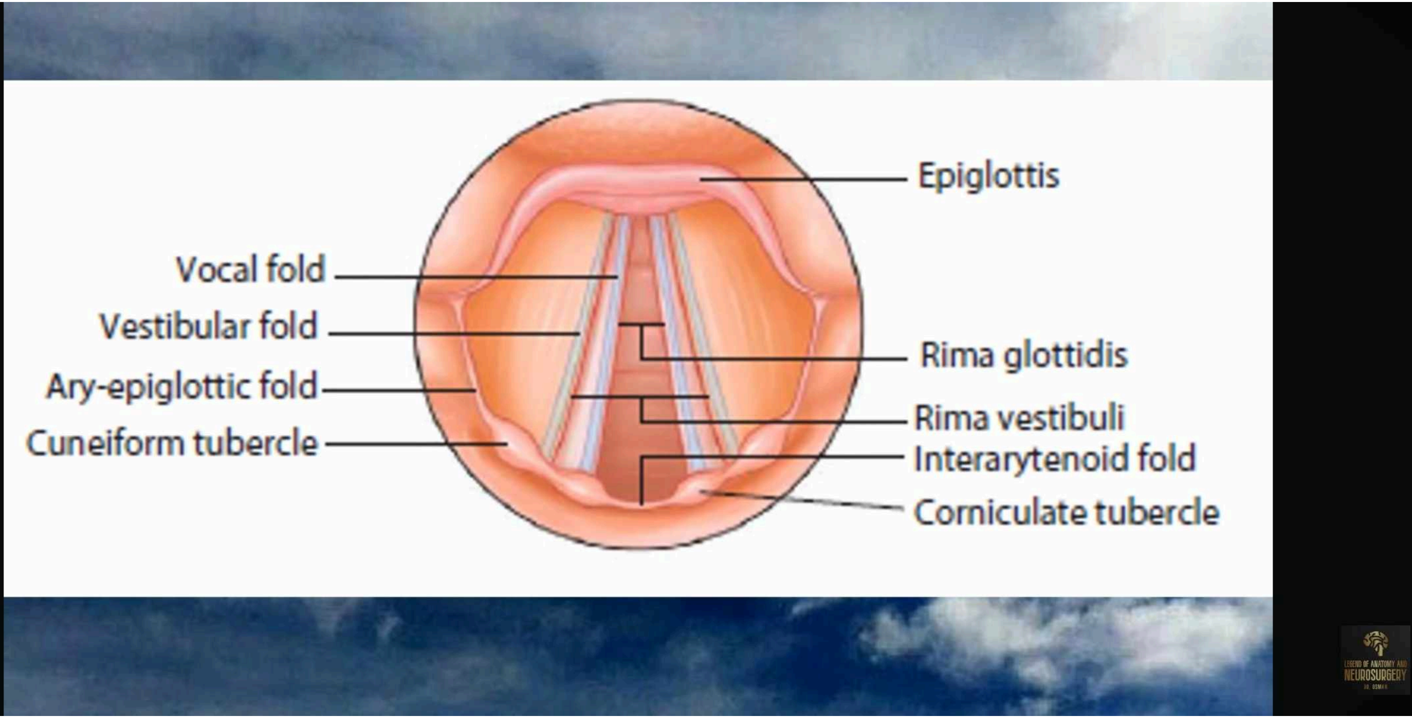

Hyoid Bone
Epiglottis
Thyrohyoid Membrane
Thyroid Cartilage
Cricothyroid Muscles
Cricothyroid Ligament
Cricoid Cartlilage
Trachea


Laryngo-
pharynx
Larynx
Trachea


Epiglottis
Hyoid Bone
Thyrohyoid Membrane
Aryepiglottic Fold
Thyroid Cartilage
Cuneiform Tubercle
Corniculate Tubercle
Transverse Arytenoid
Oblique Arytenoid
Cricoarytenoid Muscle
Trachea

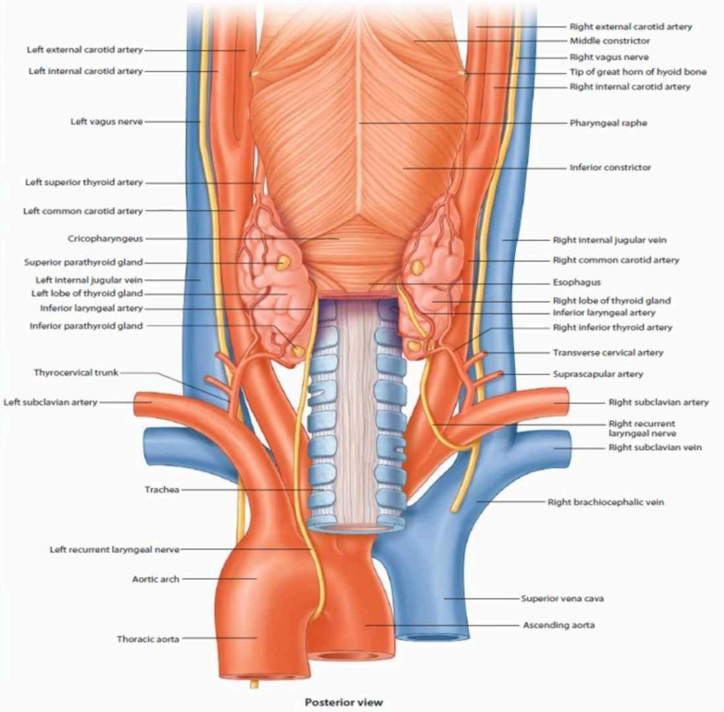


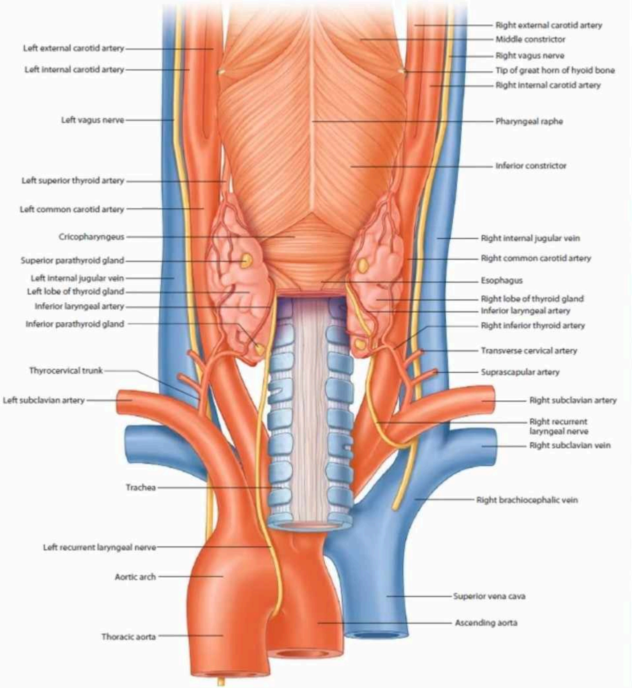
Position of larynx in the neck as it relates to the surface


Epiglottis
Vocal fold
Vestibular fold
Ary-epiglottic fold
Cuneiform tubercle
Rima glottidis
Rima vestibuli
Interarytenoid fold
Corniculate tubercle
LEGEND OF ANATOMY AND
NEUROSURGERY

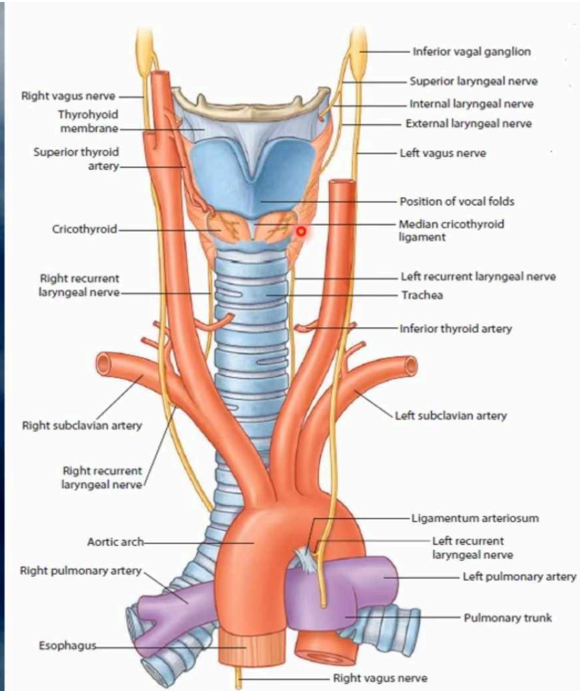
Relations

- The larynx is related to major <u>**critical**</u> structures in the neck.
- <u>**Arteries**</u>:
- <u>**Carotid arteries:**</u> **(common, external and internal).**
- <u>**Thyroid arteries**</u>**: (superior & inferior thyroid arteries).**
- <u>**Veins:**</u>
- **Jugular veins, (external & internal)**
- <u>**Nerves:**</u>
- **Laryngeal nerves: (Superior laryngeal & recurrent laryngeal).**
- **vagus nerve.**

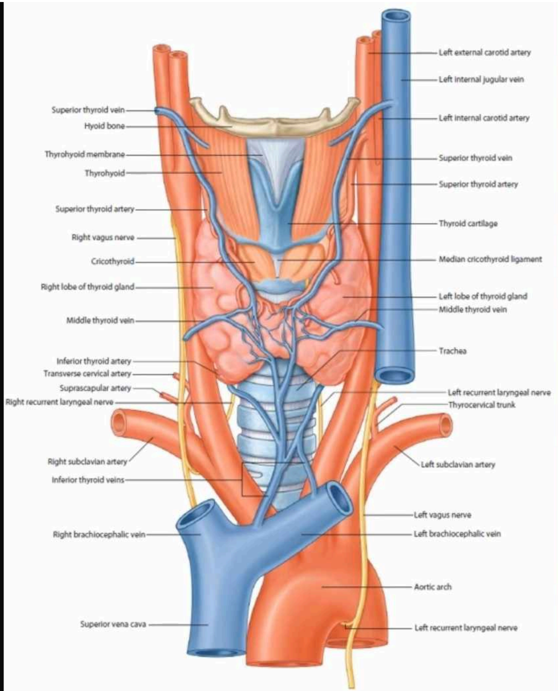




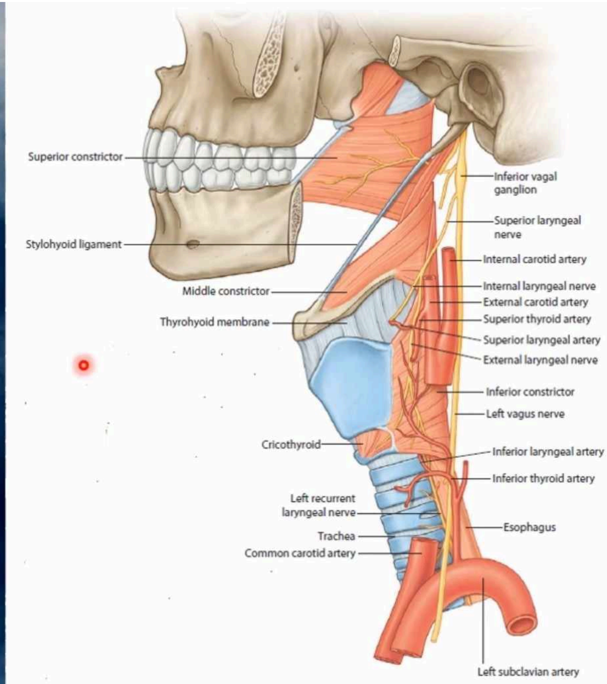
Posterior view



Anterior view



Anterior view



Lateral view

Structure

- **The larynx consists of 5 basic components**:

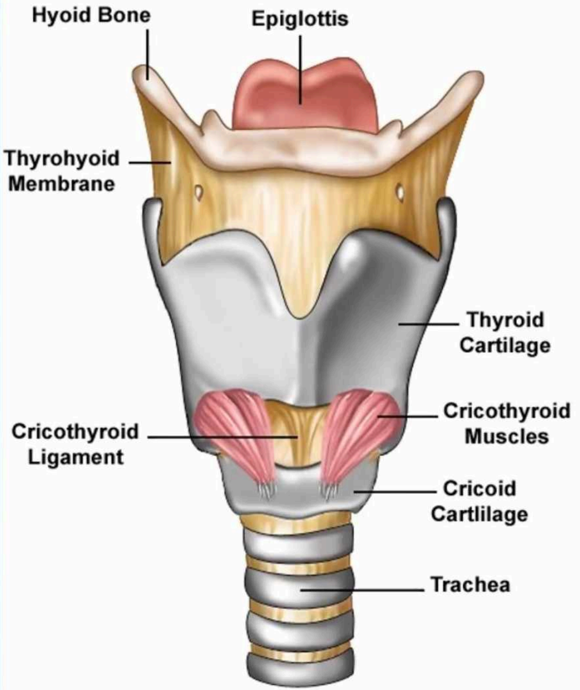
1- Cartilaginous skeleton.

2- Membranes and ligaments.

3- joints

4- Muscles (Intrinsic & extrinsic muscles).

5- Mucosal lining.



The Cartilages

- <u>**The cartilaginous skeleton**</u> is composed of:

  1. **Thyroid**
  2. **Cricoid**
  3. **Epiglottis**

  3 Single

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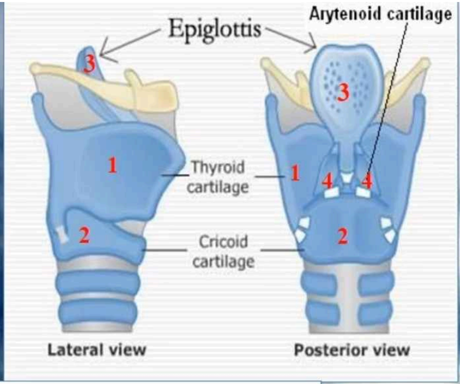
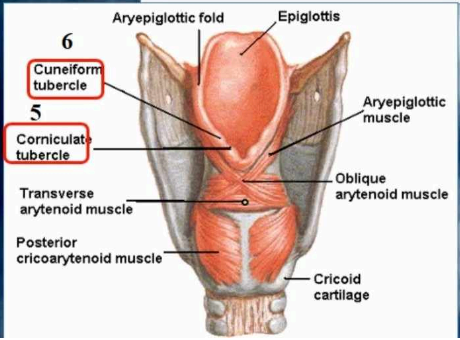
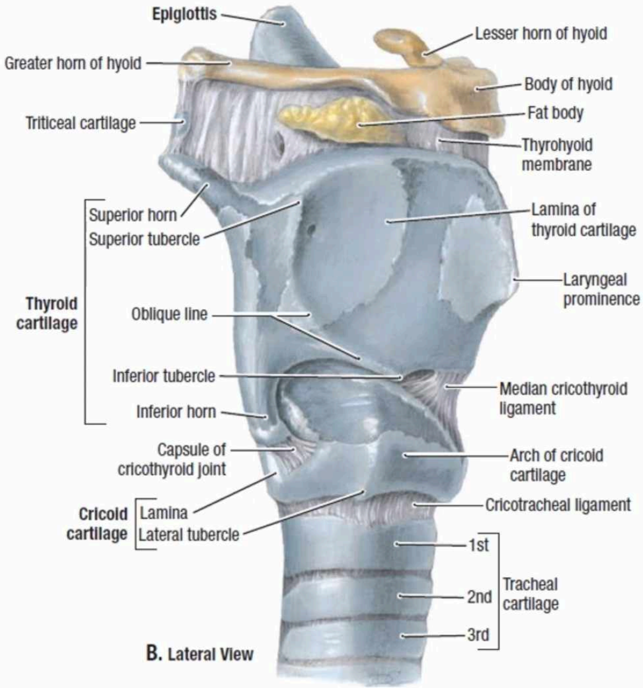
  4. **Arytenoid**
  5. **Corniculate**
  6. **Cuneiform**

  3 Paired

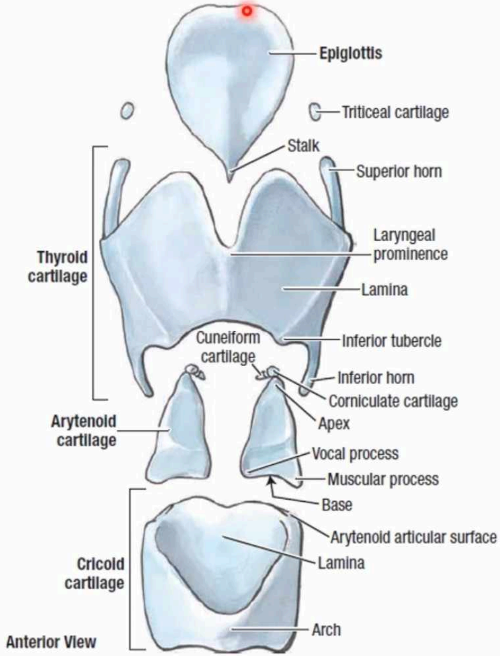
- All the cartilages, are **hyaline** except the **epiglottis** which is <u>Elastic</u> cartilage.
- <u>The cartilages are:</u>
  - **Connected by joints, membranes & ligaments.**
  - **Moved by muscles.**







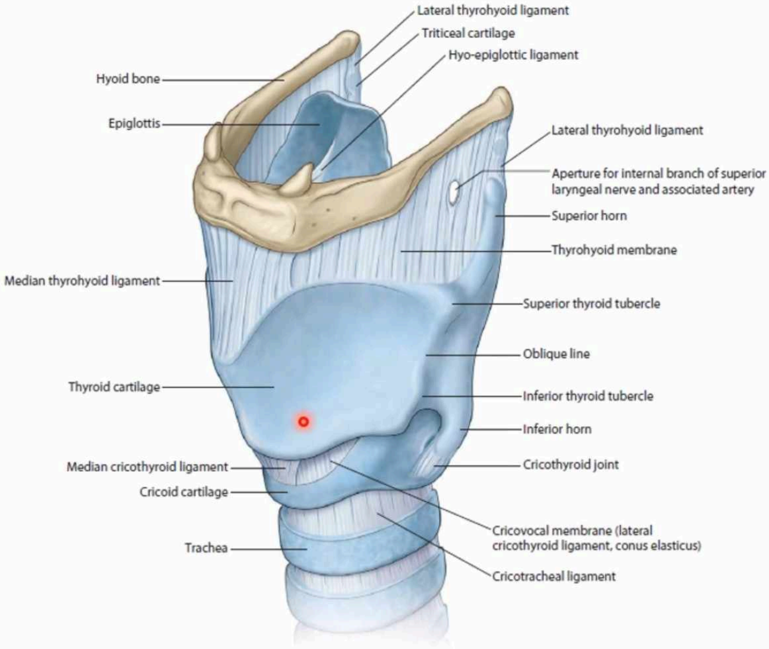
B. Lateral View



Anterior View

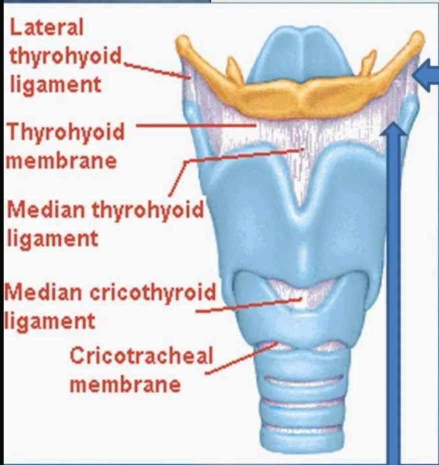
CARTILAGES OF LARYNGEAL SKELETON

Scanned with AnyScanner

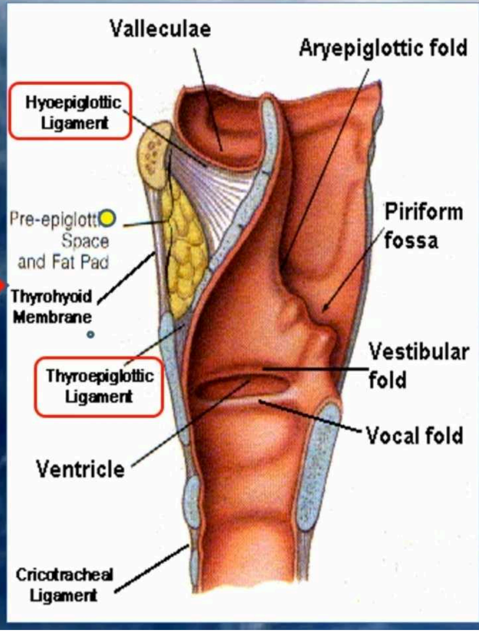


External features of the larynx (anterolateral view)

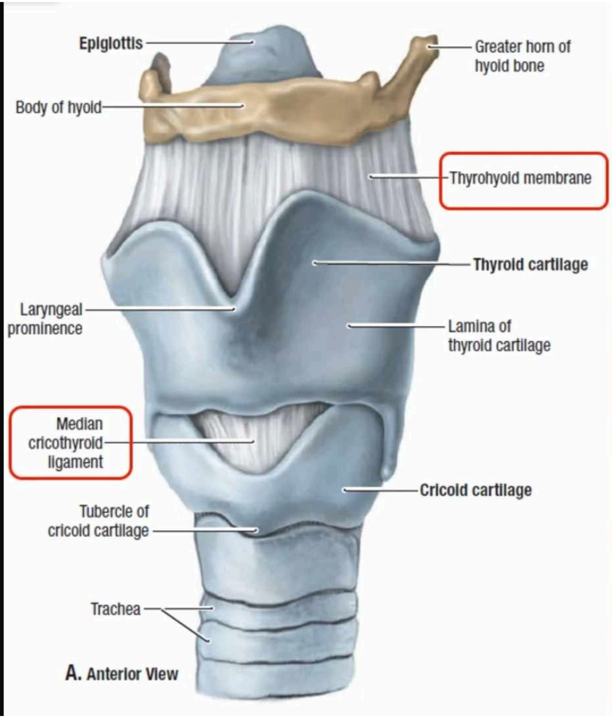
MEMBRANES & LIGAMENTS

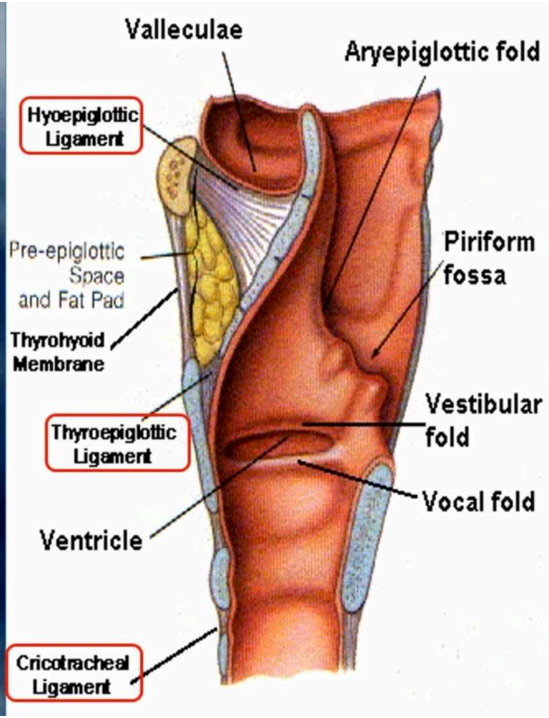


- Thyrohyoid membrane.
- Cricothyroid membrane.
- Cricotracheal membrane
- **Hyoepiglottic ligament.**
- **Thyroepiglottic ligament**



The **thyrohyoid membrane** is thickened in the median plane to form **median thyrohyoid ligament** and on both sides to form **lateral thyrohyoid ligaments.**





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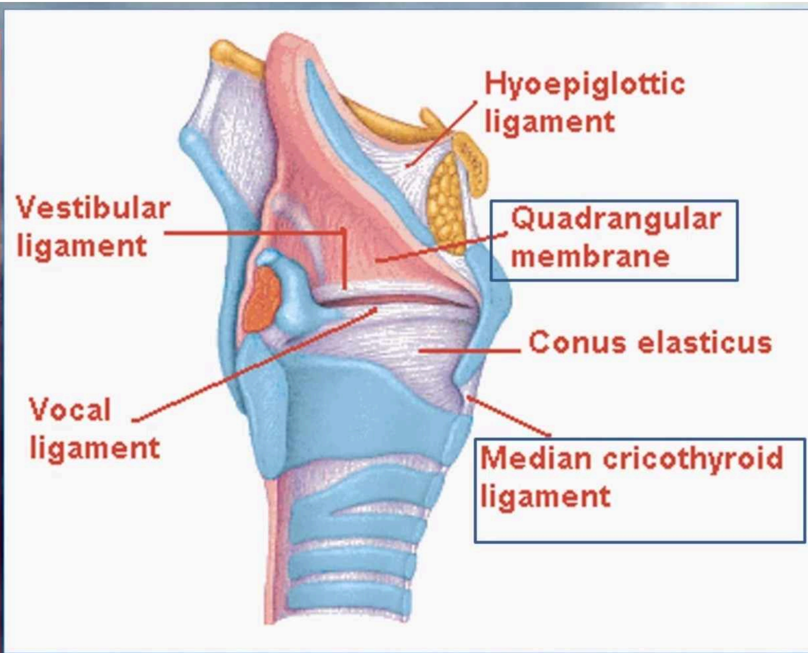
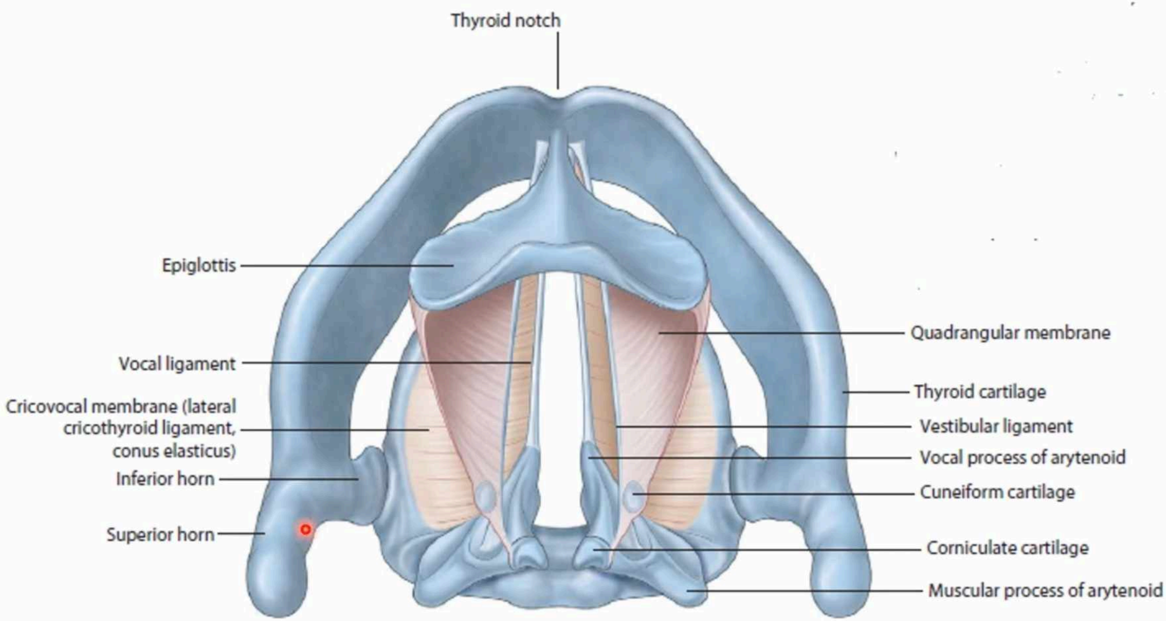
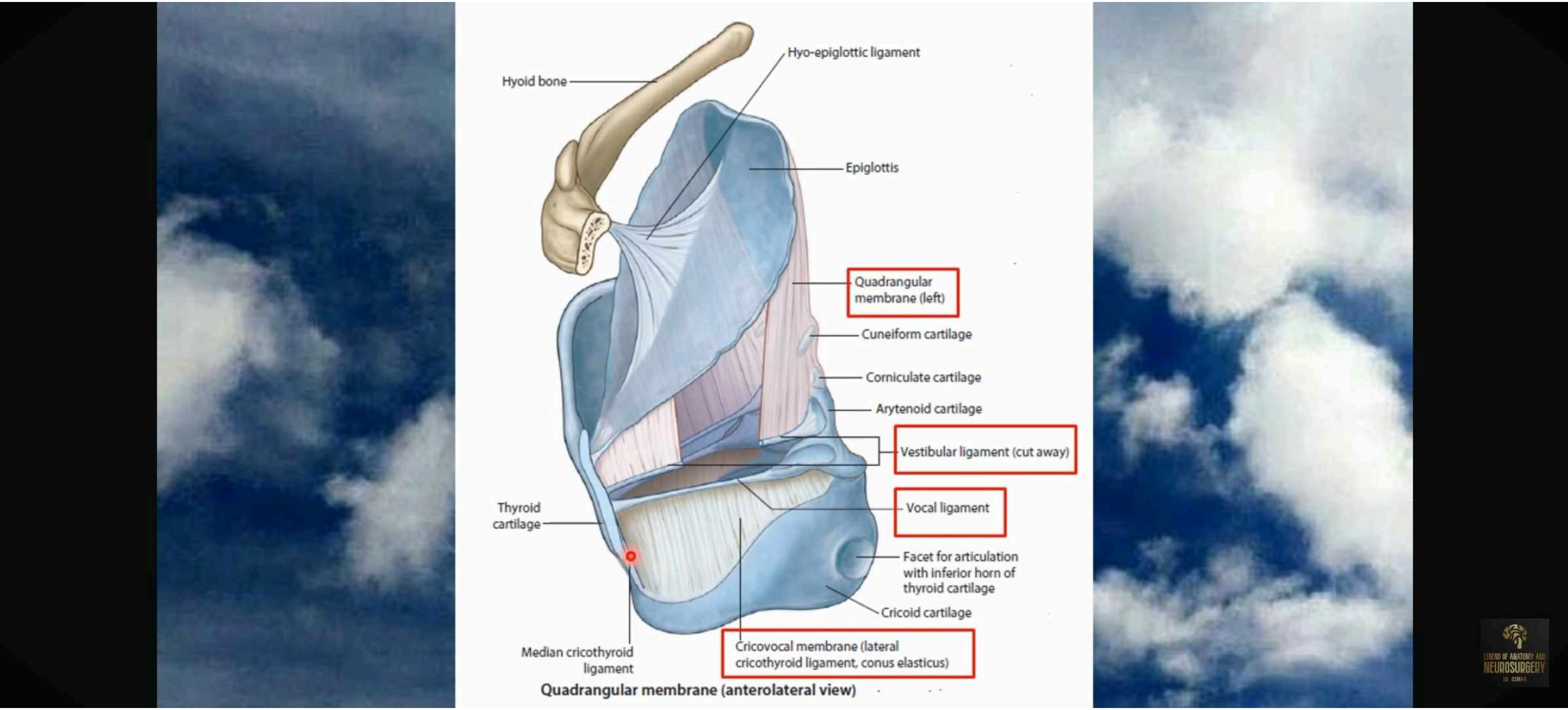
- **Quadrangular membrane:**
  - **Or aryepiglottic membrane,**
  - It extends between the arytenoid and epiglottis.
  - Its **lower free margin** forms the **vestibular ligament which forms** the **vestibular fold (false vocal cord).**
- **Cricothyroid membrane (conus elasticus):**
  - Its lower margin is attached to the upper border of cricoid cartilage.
  - **Upper free margin** forms **Vocal ligament which forms the true vocal cord**





Fibro-elastic membrane of the larynx (superior view)

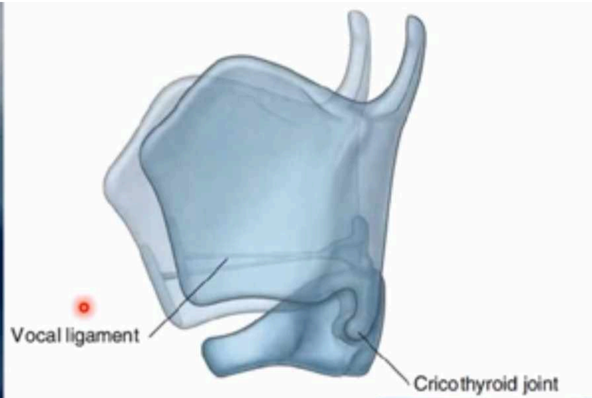


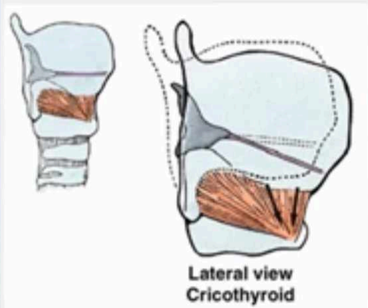
Quadrangular membrane (anterolateral view)

Joints

The cricothyroid joint

- It is between
  - **the inferior horn of the thyroid cartilage.** (and)
  - **the facet on the side of the arch of the cricoid,**
- It is synovial.
- Because the vocal ligaments pass between the posterior aspect of the thyroid angle and the arytenoid cartilages.
- That sit on the lamina of cricoid cartilage, forward movement and downward rotation of the thyroid cartilage on the of cricoid cartilage effectively lengthens and puts tension on the vocal ligaments.
- The **recurrent laryngeal nerve** lies immediately **behind** this joint.

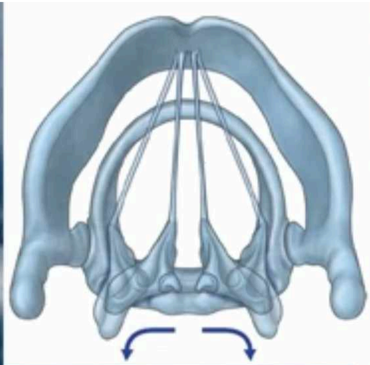


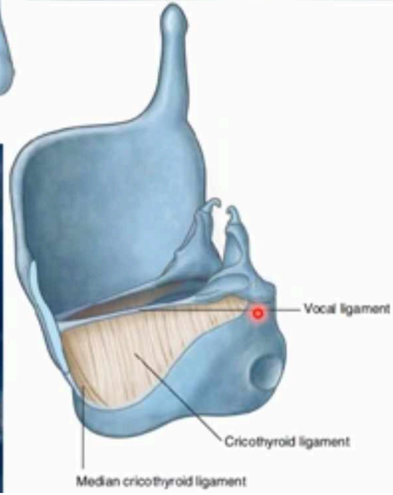


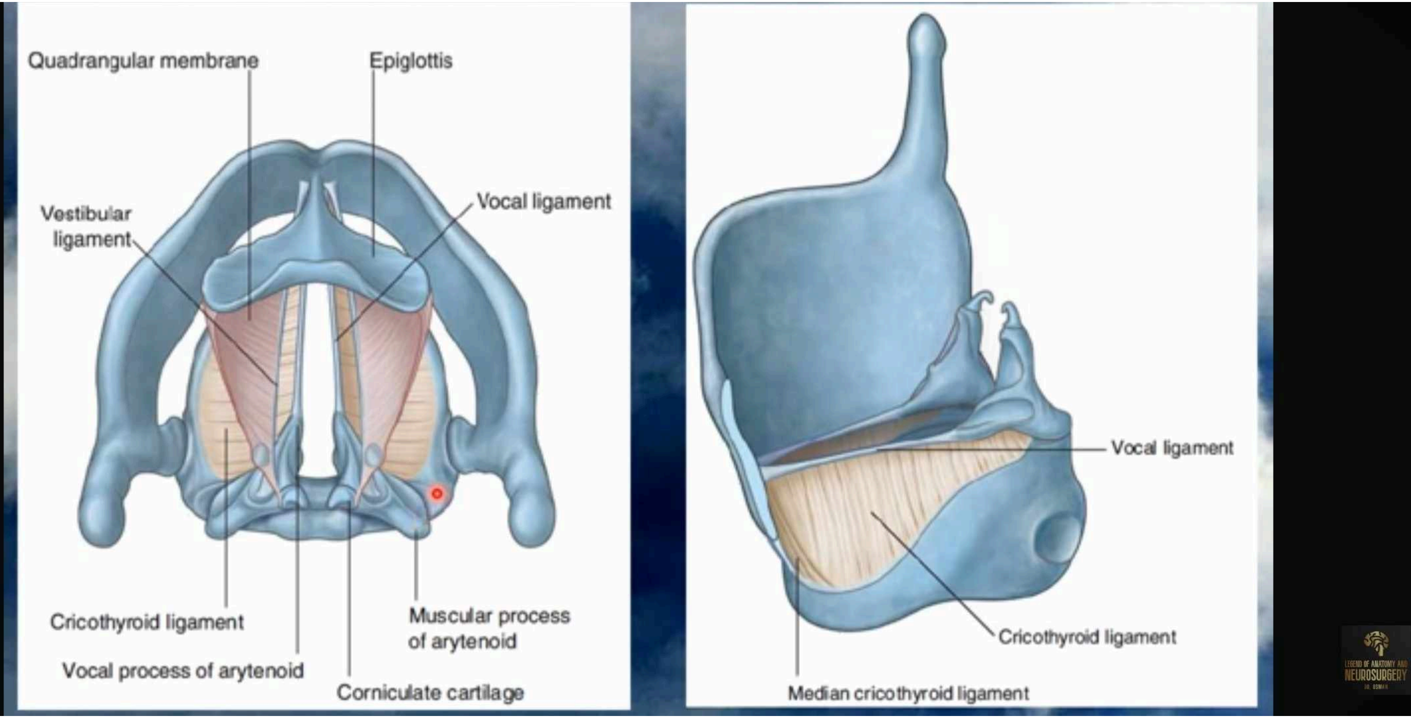
Joints

The **cricoarytenoid joint**

- The **cricoarytenoid joint**
- **It** is also synovial.
- It is between
  - **articular facets on the superiolateral surfaces of the cricoid cartilage.** (and)
  - **the bases of the arytenoid cartilages.**
- This gliding of the arytenoids opens the gap between the vocal folds (the rima of the glottis) in the shape of a V; rotation opens the glottis in the shape of a diamond.






Quadrangular membrane
Epiglottis
Vestibular ligament
Vocal ligament
Cricothyroid ligament
Vocal process of arytenoid
Corniculate cartilage
Muscular process of arytenoid
Vocal ligament
Cricothyroid ligament
Median cricothyroid ligament

Mucous Membrane

- The cavity is lined with **ciliated columnar epithelium** except the surface of the vocal cords.
- The surface of **vocal folds**, is covered with **stratified squamous epithelium** because of **exposure to continuous trauma during phonation.**
- It contains many **mucous glands**, more numerous in the region of the saccule (**for lubrication of vocal folds**).

Muscles

Laryngeal muscles are

divided into two groups:

- **Extrinsic muscles:** subdivided into two groups:
  - Elevators of the larynx.
  - Depressors of the larynx.
- **Intrinsic muscles:** subdivided into two groups:
  - Muscles **controlling the laryngeal inlet.**
  - Muscles **controlling the movements of the vocal cords.**

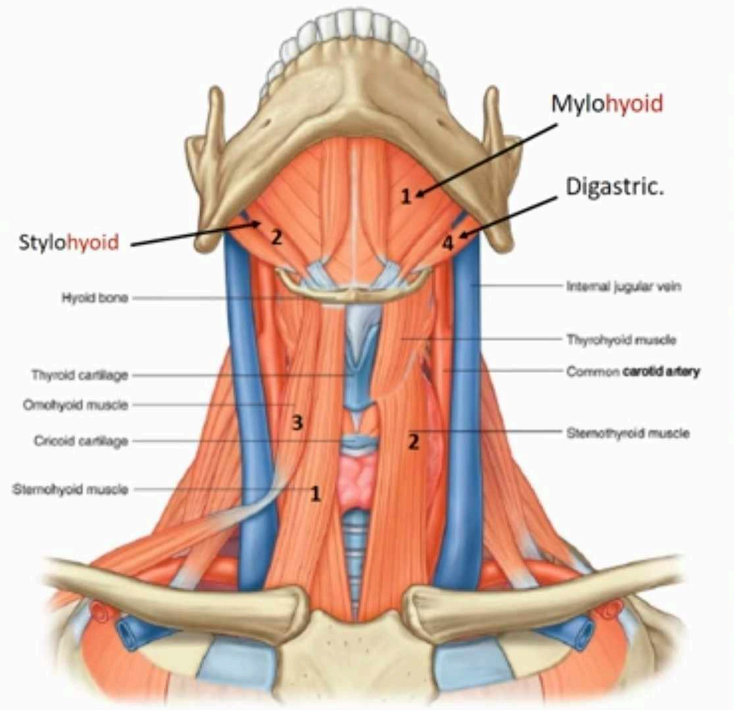
Extrinsic muscles of Larynx

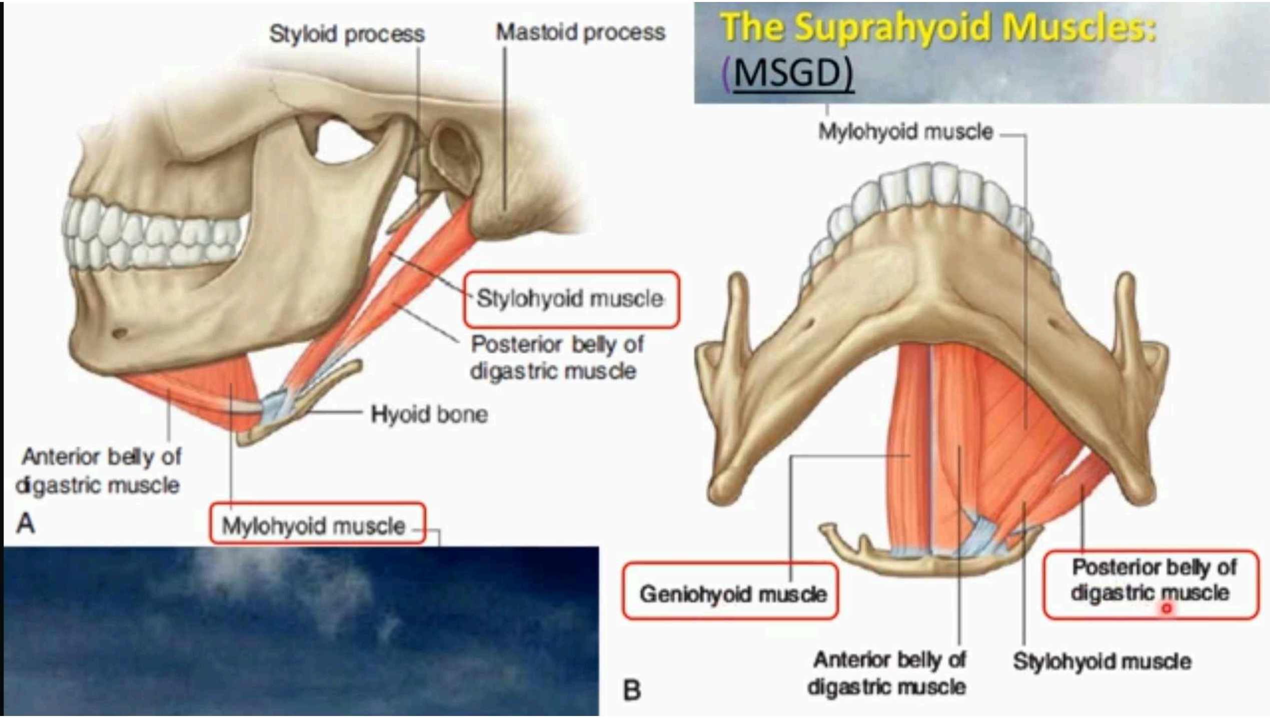
Elevators of the Larynx

- A- The Suprahyoid Muscles: (MSGD)
  1. Mylohyoid.
  2. Stylohyoid.
  3. Geniohyoid.
  4. Digastric.
- B- The Longitudinal Muscles of the Pharynx:
  - Stylopharyngeus.
  - Salpingopharyngeus.
  - Palatopharyngeus.

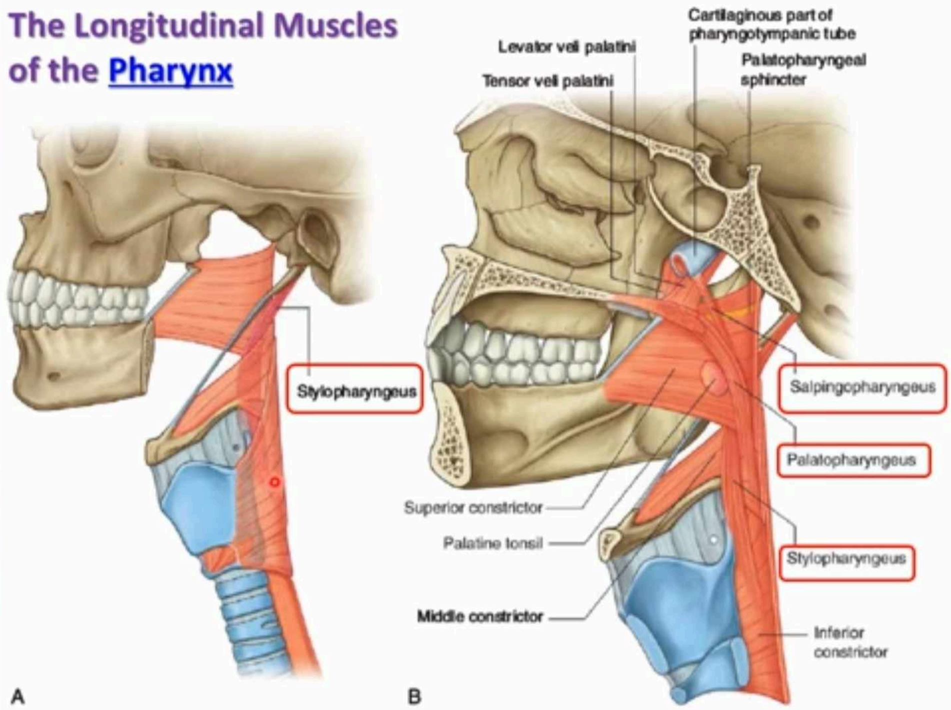
Depressors of the Larynx

- The Infrahyoid Muscles:
  1. Sternohyoid.
  2. Sternothyroid.
  3. Omohyoid.




The Suprahyoid Muscles:
(MSGD)
Styloid process
Mastoid process
Stylohyoid muscle
Posterior belly of digastric muscle
Hyoid bone
Anterior belly of digastric muscle
Mylohyoid muscle
A
Mylohyoid muscle
Geniohyoid muscle
Posterior belly of digastric muscle
Anterior belly of digastric muscle
Stylohyoid muscle
B

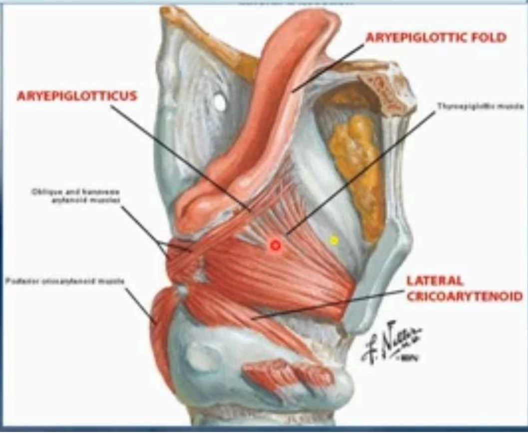
The Longitudinal Muscles of the Pharynx

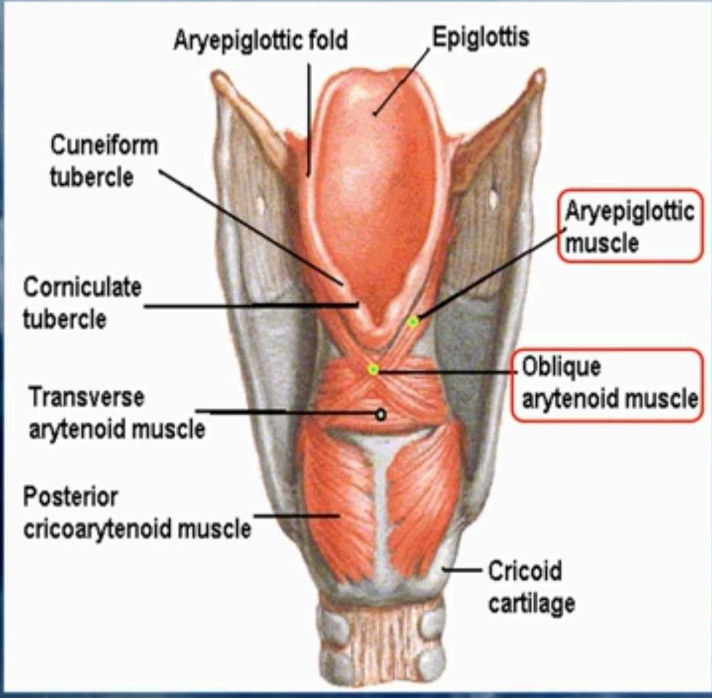


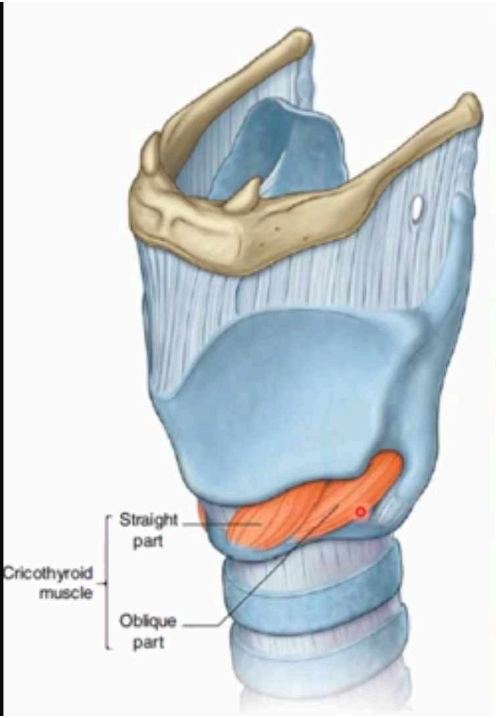
Intrinsic muscles of Larynx

Muscles Controlling the Laryngeal Inlet

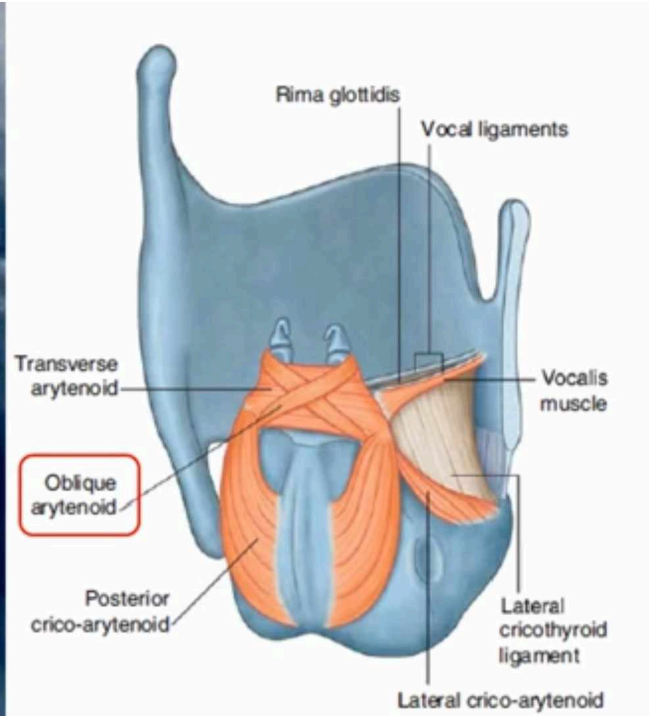
- Oblique arytenoid.
- Aryepiglottic muscle.



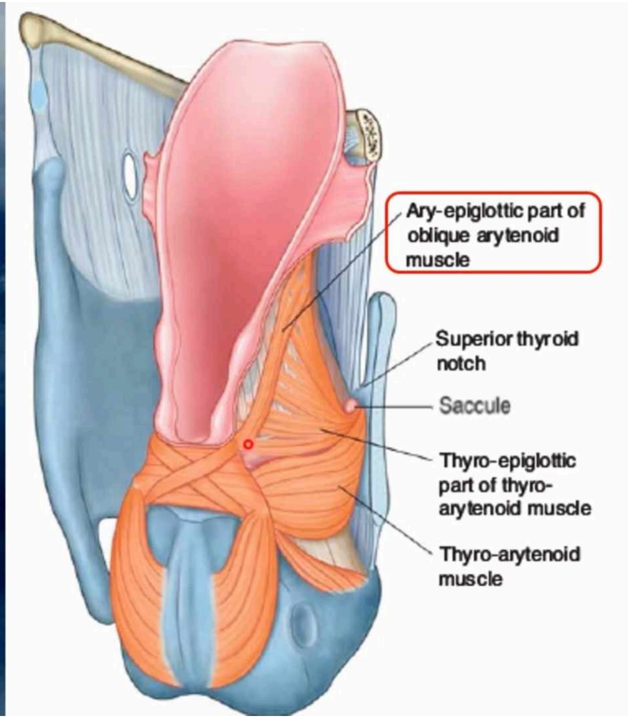




Straight part
Cricothyroid muscle
Oblique part

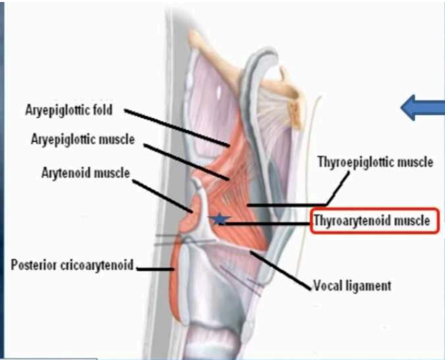


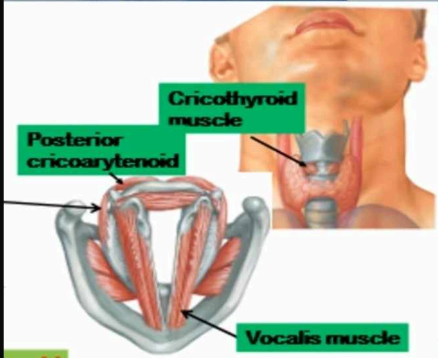
Rima glottidis
Vocal ligaments
Transverse arytenoid
Vocalis muscle
Oblique arytenoid
Posterior crico-arytenoid
Lateral cricothyroid ligament
Lateral crico-arytenoid


Ary-epiglottic part of oblique arytenoid muscle
Superior thyroid notch
Saccule
Thyro-epiglottic part of thyro-arytenoid muscle
Thyro-arytenoid muscle

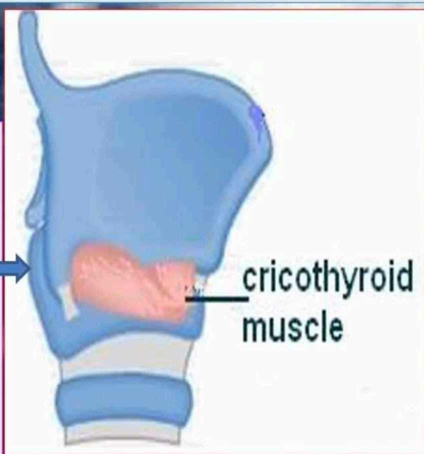
Muscles controlling the vocal cords

- **Muscle decreasing the Length & Tension of Vocal Cords** (relax vocal cords).
  - **Thyroarytenoid (vocalis).**



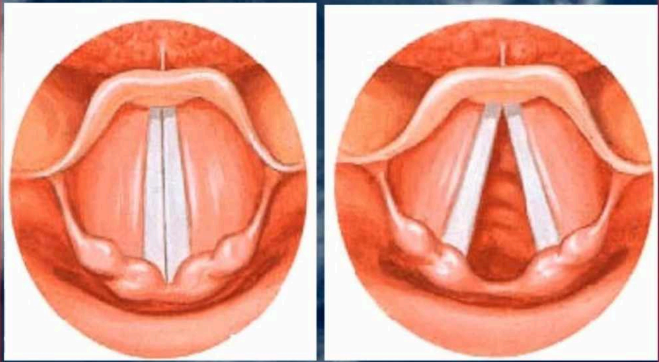


- **Muscle increasing the Length & Tension of Vocal Cords.**
  - **Cricothyroid.**
  - NB. It is the **only intrinsic** muscle which **found outside the larynx.**



Muscles controlling the vocal cords

Movements of the Vocal Cords



Adduction **Abduction**

Adductors (close rima glottis)

- Lateral cricoarytenoid.
- Transverse arytenoid.

Abductor (open rima glottis)

- Posterior cricoarytenoid.